

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Solution Architecture

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

ServiceNow Script-Controlled ACL Architecture Diagram

PRESENTATION / CLIENT LAYER - ServiceNow Next Experience UI / Classic UI Portal - Form View (u_institution_details) List View (u_institution_details_list) - Client Interactions: View Records, 'New' UI Action, In-line List Edit, 'Delete' Dropdown Option
↕ HTTPS / REST / GlideRequest Session ↕
SECURITY & GATEWAY LAYER (SERVICE SECURITY AUDIT) - Elevated Role Privilege Engine (security_admin) - Role-Based Impersonation Framework (Testing Admin vs EEE User vs Non-Role User perspectives) - Session Context & Authentication Token Validation
↕ Authorization Execution Pipeline ↕
APPLICATION & SCRIPT-CONTROLLED ACL ENGINE LAYER - Access Control List (ACL) Evaluator: Type = Record, Name = u_institution_details 4 Distinct CRUD ACL Pipelines: - READ ACL: Requires 'bb1' + Condition (Branch == 'EEE') + Script: if (gs.hasRole('bb1') gs.hasRole('admin')) return true; - CREATE ACL: Requires role 'bb2' to display & trigger 'New' button - WRITE ACL: Requires role 'bb3' to permit field and record modifications - DELETE ACL: Requires role 'bb4' to allow single & multi-row record deletions
↕ GlideRecord Database Access Layer ↕
DATA / STORAGE LAYER (SERVICENOW RELATIONAL DB) - Custom Table: u_institution_details (Student Roll No, Name, Faculty, Branch [ECE/EEE/CSE], Email, Phone) - User & Security Tables: sys_user, sys_user_role (bb1, bb2, bb3, bb4), sys_user_has_role, sys_security_acl

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the ServiceNow UI list views, form layouts, and context-dependent UI actions (New, Edit, Delete) according to access evaluations.	ServiceNow Next Experience UI, AngularJS, HTML5/CSS3
Security & Gateway Layer	Manages user authentication, impersonation testing, and elevated privilege elevation (security_admin) to guard ACL modification endpoints.	ServiceNow High Security Settings, Elevated Privileges, GlideSession
Core ACL Logic Service	Evaluates server-side script-controlled access control rules	ECMAScript 2021 (ES512 mode),

(Application Layer)	against incoming transactions, roles (admin, bb1–bb4), and field values (Branch == 'EEE').	ServiceNow GlideSystem (gs.hasRole), ACL Engine
Database / Storage Layer	Persists student records with departmental attributes and maintains system security role assignments, ACL metadata, and audit logs.	ServiceNow Relational Database Schema (u_institution_details, sys_user, sys_user_has_role)